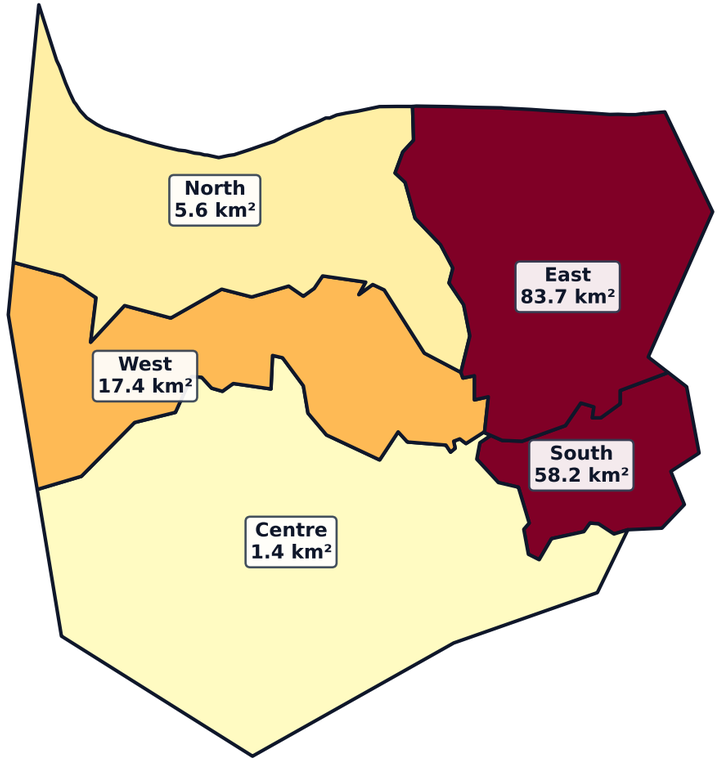
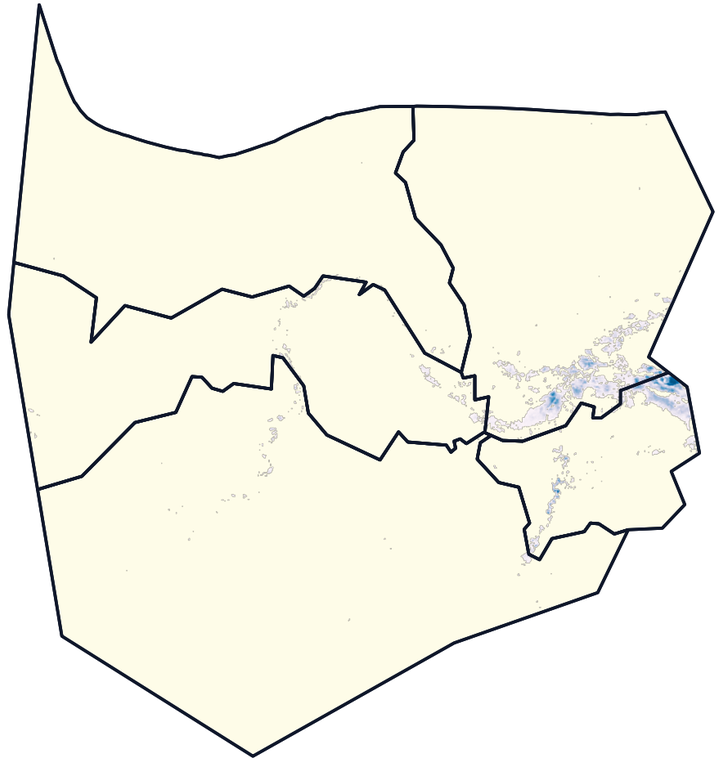
Visual 4: County-to-Grid Downscaling & Exact Mathematical Area Conservation

Step 1: Tabular County Forecast
LightGBM Multi-Head Median Output



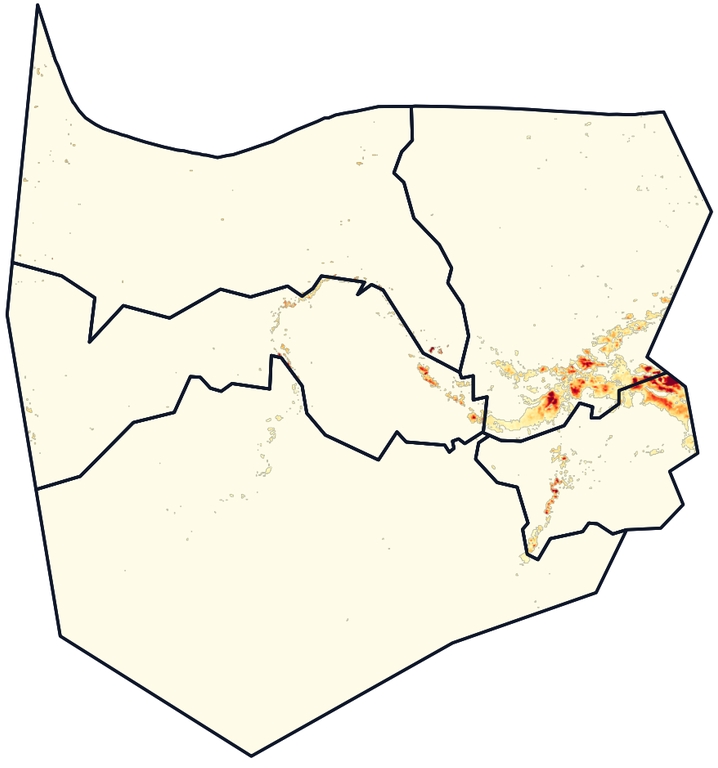
State Flood Extent Target: 166.3 km²
Aggregated county tabular quantiles (q10, q50, q90)
Uncertainty bounded at county scale

Step 2: Conditional Pixel Climatology
Historical 26-Yr Empirical Prior Mask



Conditional Probability P(pixel | month, county)
Resolves natural low-lying depressions & paleochannels
Prior informs spatial disaggregation sequence

Step 3: Downscaled Inundation Grid
Exact Area Conservation (232 m Resolution)



Target Area: 166.310 km² | Grid: 166.310 km²
Exact Conservation: Δ = 0.0000 km² (0.00% Error)
100% cell merge; zero area distortion